ISyE 7406 – Fall 2026

Project Midterm

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Team Project: Number 105

Project Title: Predicting Remittance Transfer Costs from the United States to Latin America and the Caribbean

Problem Statement

Every year, millions of Latino families in the United States send money back to relatives in Latin America and the Caribbean (LAC). These transfers are a primary source of income for receiving households. In 2023, the United States sent over $150 billion in remittances to LAC countries, making it the largest single source of remittance inflows into the region. Despite their scale and importance, the cost of sending that money remains stubbornly high. The United Nations set a target of 3% average cost per transaction under SDG 10.c[1], yet the corridors in this study average 5.44% as of 2025 Q1, with some destinations such as Cuba (12.0%) and Guyana (8.7%) far exceeding that benchmark.

The goal of this project is to understand why costs are where they are. The World Bank Remittance Prices Worldwide (RPW) database[2] tracks transaction-level pricing across hundreds of global corridors, capturing provider type, payment instrument, transfer speed, fees, exchange rate margins, and destination country for each offering. This gives us a rich, well-structured dataset, however: 16 destination countries, multiple provider categories, six payment instrument types, and nearly a decade of quarterly data, all interacting in ways a simple comparison of averages cannot fully explain.

Our approach has three parts: (1) exploratory data analysis (EDA) to identify which variables show the clearest marginal relationships with cost; (2) information-criterion and shrinkage-based feature selection to determine which predictors to carry into modeling; and (3) fitting and comparing regression models, a classification task at the 3% threshold, and corridor clustering, the work left for the final report.

[1] https://w3.unece.org/SDG/en/Indicator?id=126
[2] https://remittanceprices.worldbank.org/data-download

This midterm report covers parts 1 and 2. All analysis was conducted in Python using pandas, numpy, matplotlib, seaborn, scipy, scikit-learn, and statsmodels. Figures are described in the Appendix.

Proposed Dataset and Data Preprocessing

The dataset contains approximately 198,000 transaction-level records across global remittance corridors from 2011 Q1 to 2025 Q1. Each row represents one provider offering on one corridor in one quarter, and includes: source and destination country, provider name, provider type (Bank, Money Transfer Operator, etc.), payment instrument, access point, transfer speed, flat fee in local currency, exchange rate margin, and total cost as a percentage of the amount sent.

We subset the data to US-to-LAC corridors. The dataset contains a schema change at Q2 2016: pre-2016 records are missing payment instrument, access point, and receiving network coverage. We restrict all regression and classification modeling to post-Q2 2016 observations, yielding 12,397 rows across 16 destination countries (Brazil, Colombia, Costa Rica, Cuba, Dominican Republic, Ecuador, El Salvador, Guatemala, Guyana, Haiti, Honduras, Jamaica, Mexico, Nicaragua, Panama, Peru). The target variable is cc1 total cost %: the all-in cost of sending $200 as a percentage of the transfer amount, for the cheapest product offered by each firm in a given corridor and quarter.

Key preprocessing steps:

- **Schema harmonization:** The dataset is split across two sheets with a schema change at Q2 2016. The pre-2016 sheet is missing payment instrument, access point, and receiving network coverage. We will restrict regression and classification modeling to post-Q2 2016 observations and use the full dataset for descriptive analysis only.
- **Provider type consolidation:** "Money Transfer Operator / Building Society" (34 rows) merged into "Money Transfer Operator." Non-Bank FI (11 rows) dropped as too sparse.
- **Non-transparency flag:** Rows where cc2 total cost % is null (235 rows, 1.9%) used to engineer a binary is_non_transparent flag; excluded from regression but retained as a candidate feature.
- **Payment instrument standardization:** Multi-value entries grouped into a cleaned payment_instrument_group with 6 categories.
- **Case normalization:** speed actual and pickup method lowercased to merge near-duplicate categories.
- **Temporal engineering:** year and quarter extracted from the period field.
- **Encoding:** Categorical variables one-hot encoded with drop_first=True; final feature matrix: 12,386 rows × 42 columns.
- **Leakage exclusions:** All cc2 * fields excluded (mechanically derived from the same pricing as the target). cc1 lcu fee and inter lcu bank fx excluded as

local-currency fields not comparable across the 16 destination countries without normalization.

Proposed Methodology

Firstly, we used regression. We compare four models on the same feature set: Multiple Linear Regression (interpretable baseline), Ridge Regression (L2 regularization; handles multicollinearity from one-hot encoded destination and instrument dummies), LASSO (L1 regularization; implicit feature selection by driving low-signal coefficients to zero), and Random Forest Regression (captures nonlinear interactions between corridor, provider type, and payment method).

Then we use classification. We binarize cc1 total cost % at 3% and apply logistic regression, LDA, and Random Forest classification. We report accuracy, precision, recall, and F1-score, with the confusion matrix broken out by provider type and destination.

We then used clustering. We aggregate the 16 destination countries by mean total cost, mean FX margin, mean fee, and transparency rate, then apply k-means and hierarchical clustering to group corridors by cost profile.

Also, we use a temporal split, pre-2023 for training, 2023 onward for testing, to simulate a realistic forecasting setting and prevent data leakage across time periods.

The evaluation metrics we will use, were as follow. Regression models will be compared using RMSE (primary), MAE, and R^2. RMSE is the primary metric because it penalizes large errors more heavily, which matters given the right-skewed target and high-cost outlier corridors like Cuba. Classification models will be evaluated by precision, recall, and F1-score in addition to accuracy.

Regarding the Software we use: Python 3.x; pandas, numpy, scikit-learn, statsmodels, matplotlib, seaborn, scipy.

Analysis and Results

Target Variable: The target cc1 total cost % has a mean of 5.34%, a median of 4.71%, and a standard deviation of 3.03%, with a range of −4.09% to 34.0%. It is right-skewed (skewness = 1.88, excess kurtosis = 6.33), driven by a small number of high-cost providers in outlier corridors. The 25th percentile (3.50%) sits just above the SDG 3% target, meaning roughly three-quarters of all transactions in the dataset fail to meet it. The right-skewed distribution suggests plain OLS may be sensitive to outliers; Ridge/LASSO regularization and a possible log-transform of the target are worth evaluating in the modeling stage. *(Appendix, Figure 1)*

Cost by Destination Country: Destination is the single strongest driver of cost variation in the dataset. Cuba averages 12.0% and Guyana 8.9%, both more than double the SDG target. At the other end, Honduras (3.9%), Ecuador (4.0%), and El Salvador (4.0%) are close to the threshold. Mexico, which carries the largest absolute volume of US-to-LAC remittances, averages 4.6%. The spread of roughly 8 percentage points between the cheapest and most expensive corridors far exceeds the differences attributable to firm type or payment instrument. *(Appendix, Figure 2)*

Cost by Firm Type: A notable finding from the EDA is that **banks are not cheaper than money transfer operators (MTOs)**. Banks show a higher mean cost (5.19%) than MTOs (5.48%), and both sit well above the 3% SDG target. The difference is small in absolute terms, about 0.29 percentage points, or under $0.60 on a $200 transfer, and the more important observation is that neither category consistently reaches the affordability threshold. MTOs dominate the market by volume (10,673 of 12,386 modeling rows, 86%). The two sparse categories (MTO/Building Society, Non-Bank FI) were consolidated or dropped as described in Section 2. *(Appendix, Figure 3)*

Cost by Payment Instrument: Payment instrument differentiates cost meaningfully. Cash-based transfers show higher and more variable costs than bank account transfers or digital instruments. Credit card transfers carry a distinct premium, likely reflecting card processing fees. This supports retaining payment_instrument_group as a predictor. *(Appendix, Figure 4)*

Cost Over Time: Mean cost has been essentially flat across 2016–2025, fluctuating between 4.96% (2020) and 5.59% (2024), with no sustained downward trend toward the SDG target. year is therefore a low-signal predictor, retained only because it is needed for the temporal train/test split. *(Appendix, Figure 5)*

Transparency Flag: We engineered a binary is_non_transparent flag using cc2 total cost %-is-null as a proxy for non-disclosure of exchange rates. The 235 flagged rows (1.9%) show a modestly lower mean cost than transparent rows (4.80% vs. 5.35%). However, the proxy is imperfect: 81% of flagged rows carry a

"send limits" disclosure note rather than an exchange-rate non-disclosure note. The flag is retained as a candidate feature with this limitation stated explicitly. *(Appendix, Figure 6)*

Correlation Structure: Among the numeric fields, cc1 fx margin shows the strongest correlation with the target (r = 0.71), which is expected since the FX margin is a direct cost component. cc1 lcu fx rate and inter lcu bank fx correlate weakly (r ≈ 0.06–0.07) and are currency-scale fields excluded from the predictor set. year correlates near zero with cost (r = 0.02), consistent with the flat time trend. All cc2 * fields show high correlation with the target by construction (r = 0.91 for cc2 total cost %) and are excluded as leakage variables. *(Appendix, Fig.7)*

Feature Selection: Feature selection used three complementary methods: information-criterion-based forward stepwise selection (AIC/BIC), and shrinkage-based selection via Ridge and LASSO regression.

Forward stepwise selection (AIC/BIC): We ran a greedy forward search adding one categorical block or numeric feature at a time, stopping when BIC no longer improved. The search selected 9 of 10 candidate blocks in the following order: cc1 fx margin, pickup method, payment_instrument_group, destination_name, access point, receiving network coverage, speed actual, year, is_non_transparent. The only block not selected by BIC was firm_type_clean, which did not improve BIC further once the other predictors were in the model. *(Appendix, Figure 8)*

LASSO (LassoCV, 5-fold CV): At the cross-validated alpha (0.001), LASSO zeroed out 6 of 42 encoded features — sparse destination and access-point dummies. The largest surviving coefficients were pickup method_own/partner bank account (9.02), destination_name_Cuba (5.60), access point_Bank branch,Internet (3.87), pickup method_bank account (2.93), payment_instrument_group_Credit Card (2.46), and cc1 fx margin (2.22). firm_type_clean_Money Transfer Operator survived with a small positive coefficient (rank 34 of 36 surviving features by magnitude). *(Appendix, Figures 9–10)*

Ridge (RidgeCV, 5-fold CV): RidgeCV selected alpha = 1.15 and shrinks all 42 coefficients smoothly toward zero without zeroing any out, controlling variance from the high-dimensional one-hot dummy block. It will serve as a comparison against LASSO in the regression modeling stage.

Convergence: Both the IC-based and shrinkage-based methods point to the same dominant predictors: destination country, payment instrument/pickup method, and FX margin. This convergent evidence gives confidence the selected feature set is robust.

firm_type_clean is a deliberate exception: excluded by forward BIC selection and low-ranked by LASSO, but retained on domain reasoning. The EDA shows a real cost gap between banks and MTOs; the methods absorb and dilute this effect once destination and payment instrument are already in the model, since firm type and

corridor/payment method are correlated in this dataset. This is flagged as a judgment call.

PCA (numeric block): PCA on the three numeric candidates (cc1 fx margin, year, is_non_transparent) confirmed they are near-orthogonal — all three components are needed to explain ~100% of variance, so dimensionality reduction is not useful here. *(Appendix, Figure 11)*

Final feature set (pre-encoding source columns): destination_name, firm_type_clean, payment_instrument_group, access point, receiving network coverage, speed actual, pickup method, cc1 fx margin, year, is_non_transparent. After one-hot encoding with drop_first=True: 42 features.

Conclusions and Next Steps

The EDA and feature selection work confirms the dataset is well-suited for the planned modeling tasks. Key findings to date:

- The mean total cost of US-to-LAC remittances (5.44%) has remained well above the UN SDG 10.c target of 3% throughout 2016–2025, with no meaningful downward trend.
- Destination country is the dominant driver of cost variation, with an 8+ percentage point spread between the cheapest (Honduras, 3.9%) and most expensive (Cuba, 12.0%) corridors.
- Banks are not systematically cheaper than MTOs in this dataset, both categories average well above the SDG threshold, and the cost gap between them is under 0.3 percentage points on average.
- cc1 fx margin, payment instrument, pickup method, and destination are the strongest predictors under both information-criterion and shrinkage-based selection.
- The cc2 * fields must be excluded from all predictive models due to target leakage. cc1 lcu fee and inter lcu bank fx must be excluded as currency-scale, non-comparable predictors.

Remaining work for the final report:

1. Fit and compare regression models (Linear, Ridge, LASSO, Random Forest) on the pre-2023 training set; evaluate on the 2023+ test set using RMSE, MAE, and R^2.
2. Run the classification task at 3% and report precision, recall, and F1-score by corridor and provider type.
3. Cluster the 16 destination countries by cost profile using k-means and hierarchical clustering.
4. Plot and interpret residuals, feature importances, and LASSO coefficients in the context of the final models.
5. Write the "Lessons Learned" subsection.

Appendix

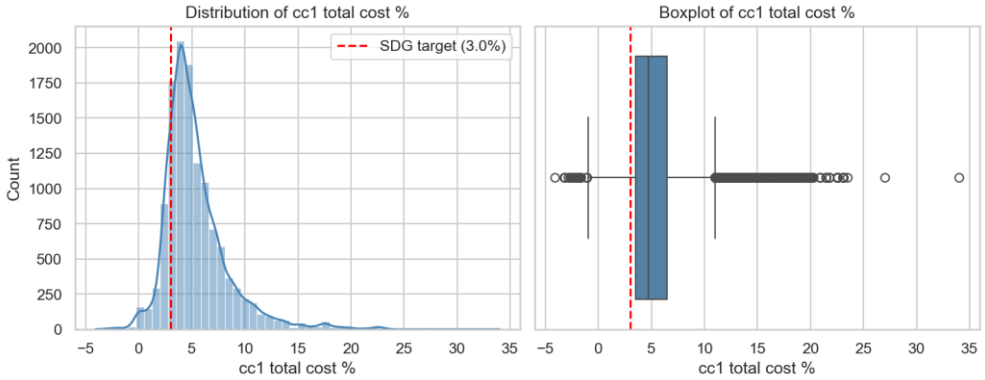


Figure 1. Distribution of `cc1 total cost %` — histogram with KDE and SDG 3% reference line; boxplot. Right-skewed, mean 5.34%, median 4.71%.

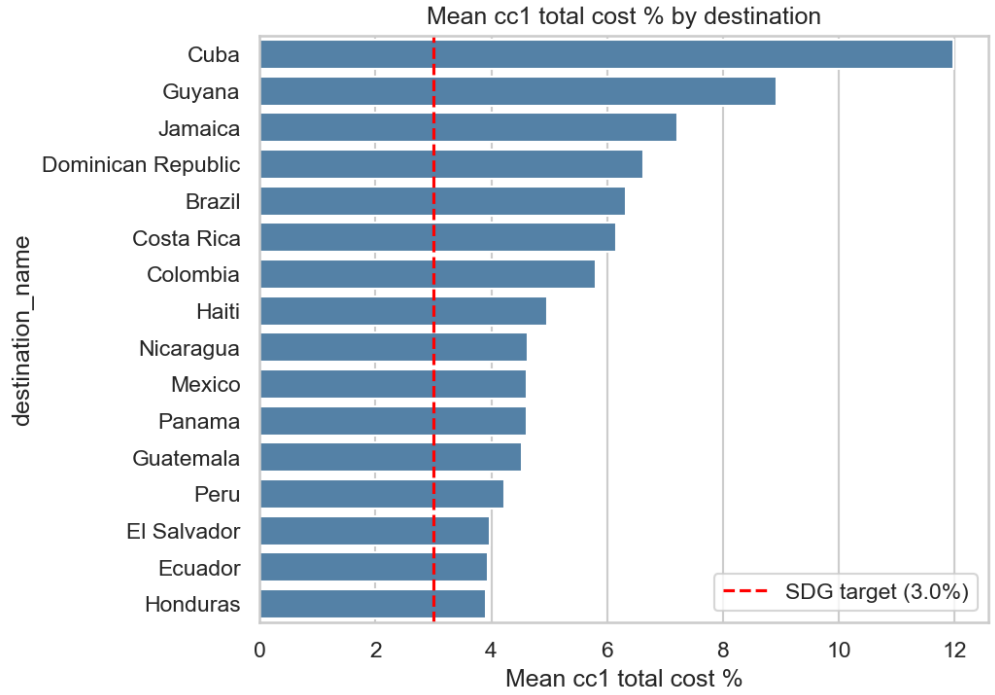


Figure 2. Mean `cc1 total cost %` by destination country — bar chart sorted descending, SDG reference line. Cuba (12.0%) and Guyana (8.9%) are outliers; Honduras (3.9%) is closest to target.

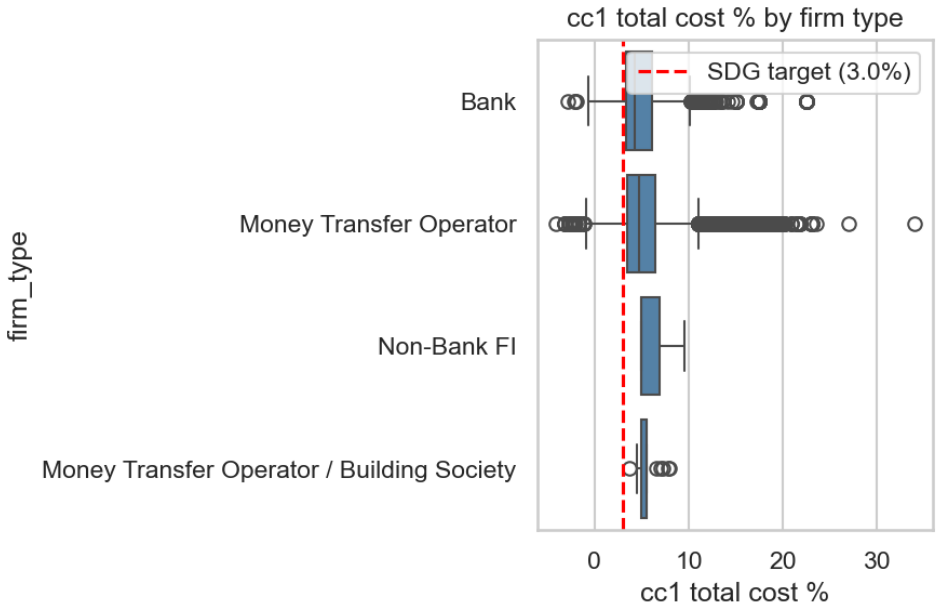


Figure 3. cc1 total cost % by firm type — boxplot. Banks show higher median cost than MTOs.

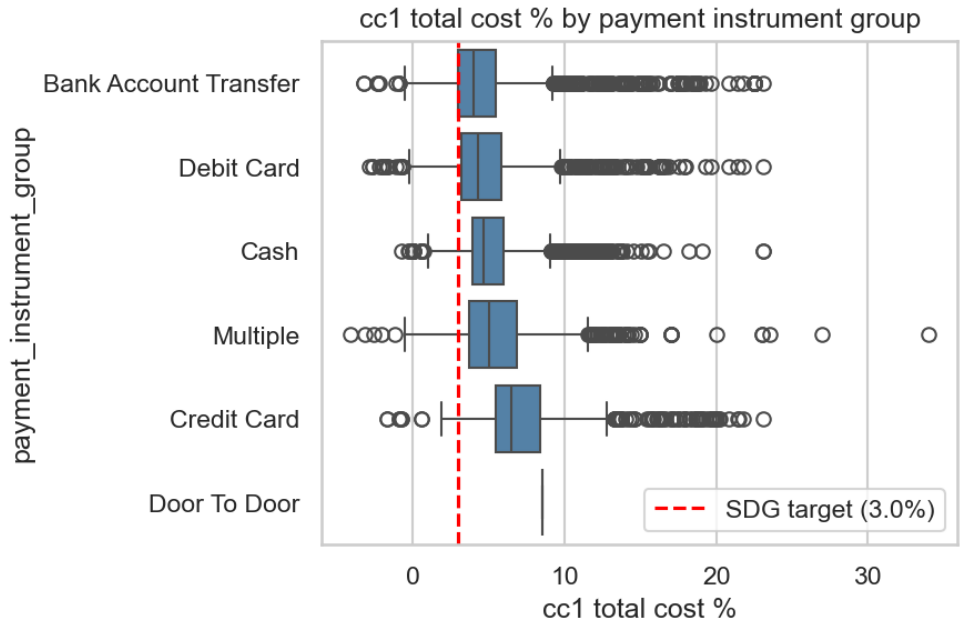


Figure 4. cc1 total cost % by payment instrument group — boxplot. Cash and credit card carry higher costs.

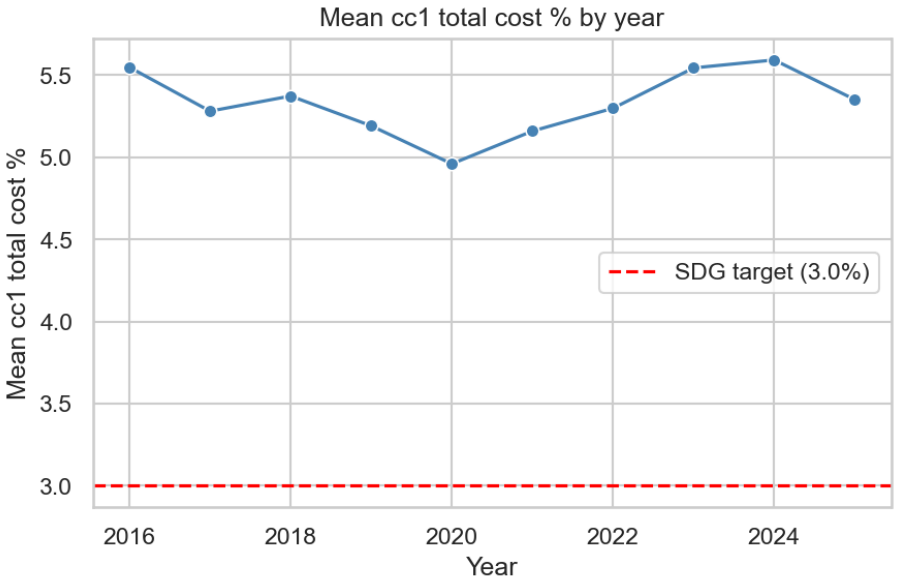


Figure 5. Mean `cc1 total cost %` by year (2016–2025) — line chart. Flat trend, never approaches 3%.

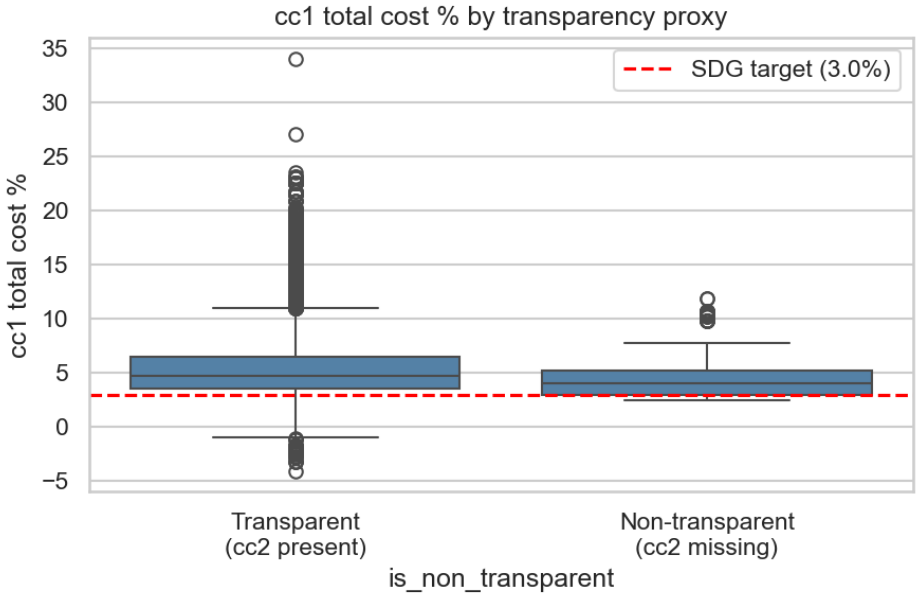
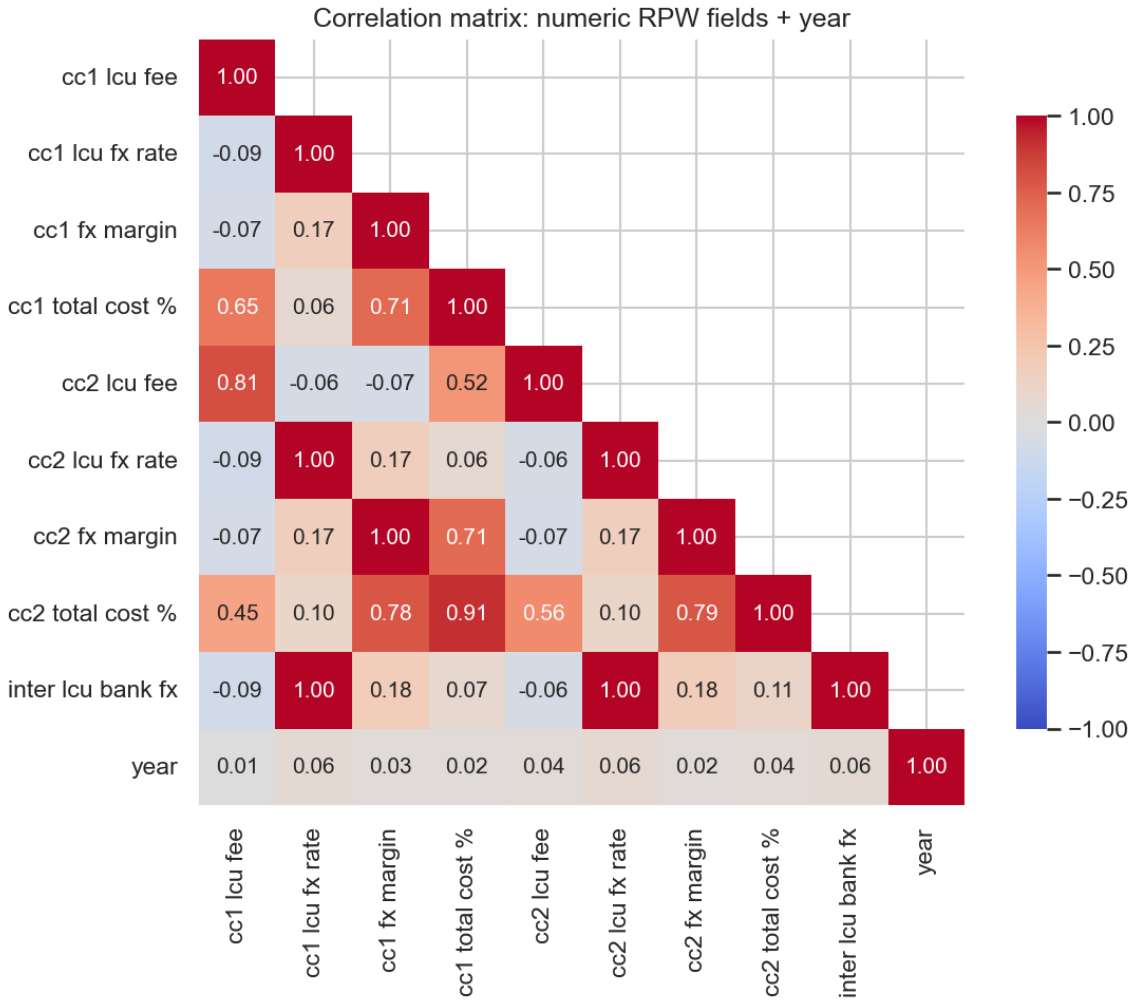


Figure 6. `cc1 total cost %` by transparency proxy (cc2 present vs. missing) — boxplot. Non-transparent rows show modestly lower costs (4.80% vs. 5.35%).



Field	cc1 lcu fee	cc1 lcu fx rate	cc1 fx margin	cc1 total cost %	cc2 lcu fee	cc2 lcu fx rate	cc2 fx margin	cc2 total cost %	inter lcu bank fx	year
cc1 lcu fee	1.0	-0.09	-0.07	0.65	0.81	-0.09	-0.07	0.45	-0.09	0.01
cc1 lcu fx rate	-0.09	1.0	0.17	0.06	-0.06	1.0	0.17	0.1	1.0	0.06
cc1 fx margin	-0.07	0.17	1.0	0.71	-0.07	0.17	1.0	0.78	0.18	0.03
cc1 total cost %	0.65	0.06	0.71	1.0	0.52	0.06	0.71	0.91	0.07	0.02
cc2 lcu fee	0.81	-0.06	-0.07	0.52	1.0	-0.06	-0.07	0.56	-0.06	0.04
cc2 lcu fx	-0.09	1.0	0.17	0.06	-0.06	1.0	0.17	0.1	1.0	0.06

rate										
cc2 fx margin	-0.07	0.17	1.0	0.71	-0.07	0.17	1.0	0.79	0.18	0.02
cc2 total cost %	0.45	0.1	0.78	0.91	0.56	0.1	0.79	1.0	0.11	0.04
inter lcu bank fx	-0.09	1.0	0.18	0.07	-0.06	1.0	0.18	0.11	1.0	0.06
year	0.01	0.06	0.03	0.02	0.04	0.06	0.02	0.04	0.06	1.0

Figure 7. Pearson correlation matrix — numeric RPW fields plus year. `cc1 fx margin` has the strongest independent correlation with the target (r = 0.71).

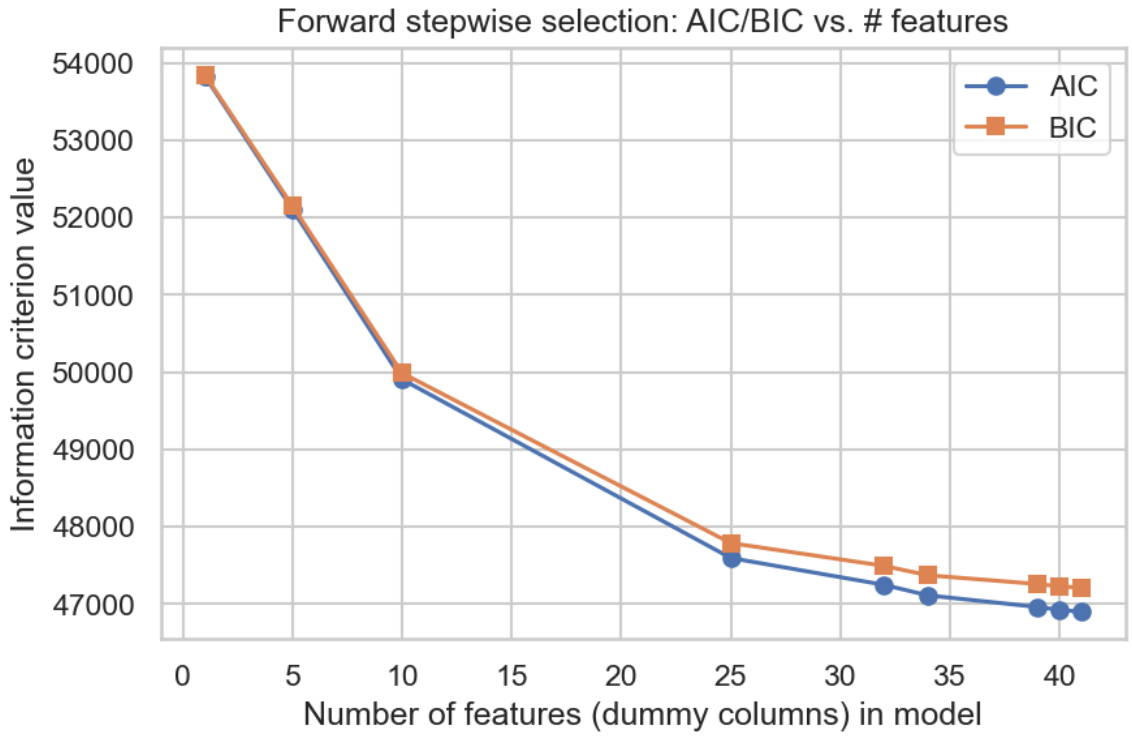


Figure 8. Forward stepwise selection — AIC and BIC vs. number of features added. BIC curve flattens after destination and instrument blocks are included.

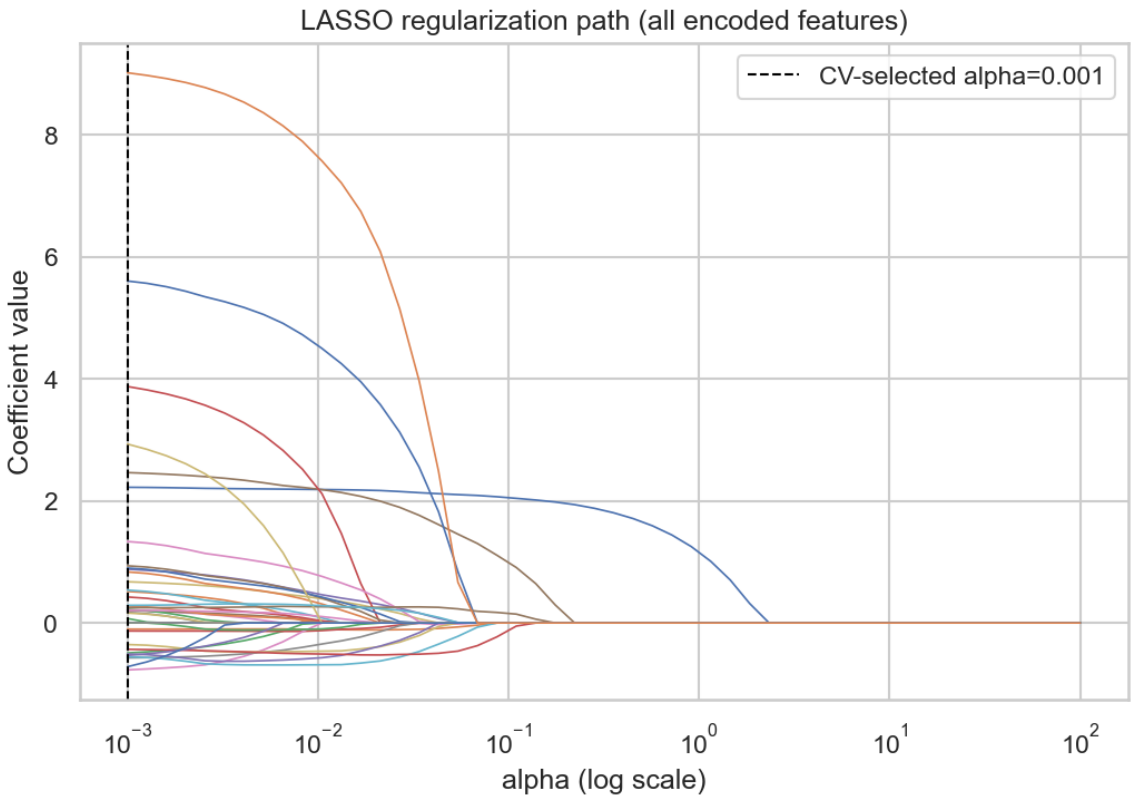


Figure 9. LASSO regularization path — all 42 encoded features, CV-selected alpha (0.001) marked.

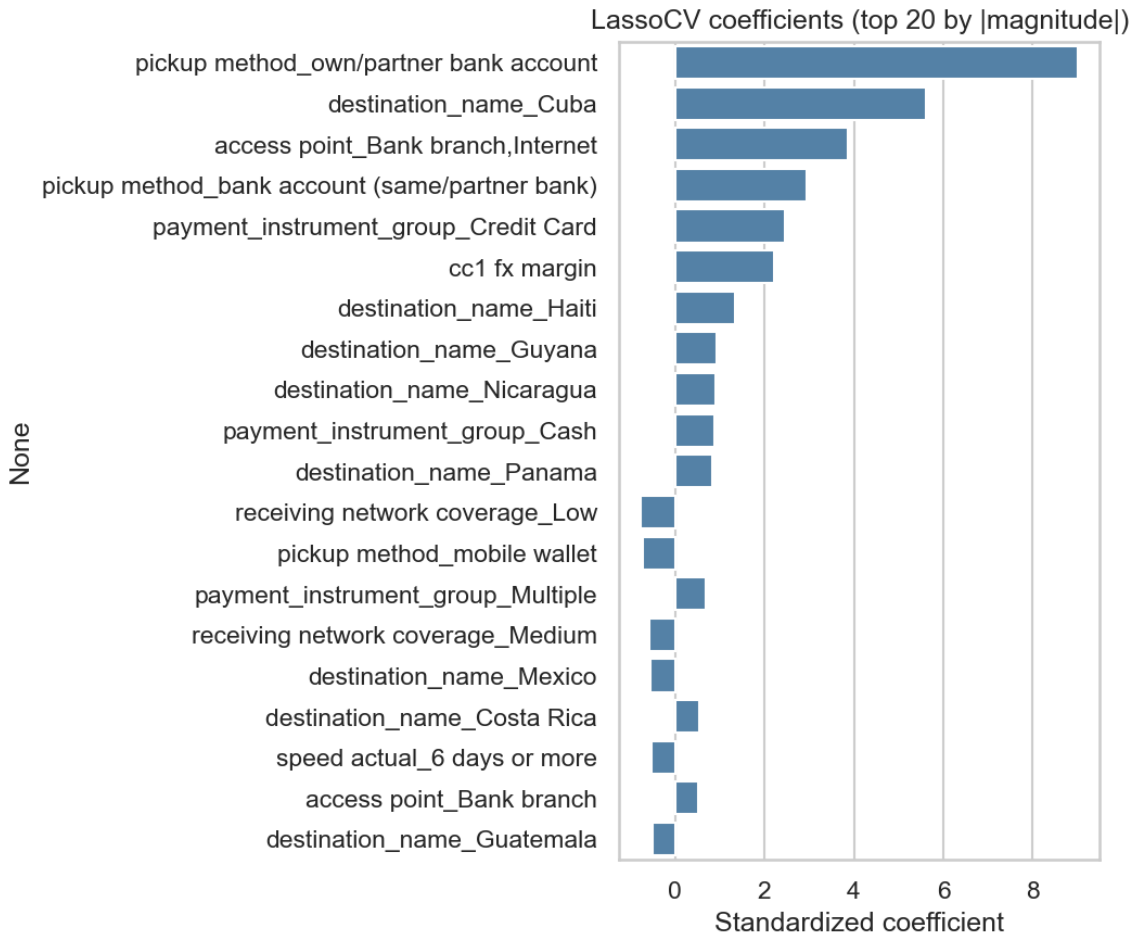


Figure 10. LassoCV coefficients — top 20 by absolute magnitude. pickup method_own/partner bank account and destination_name_Cuba are the largest.

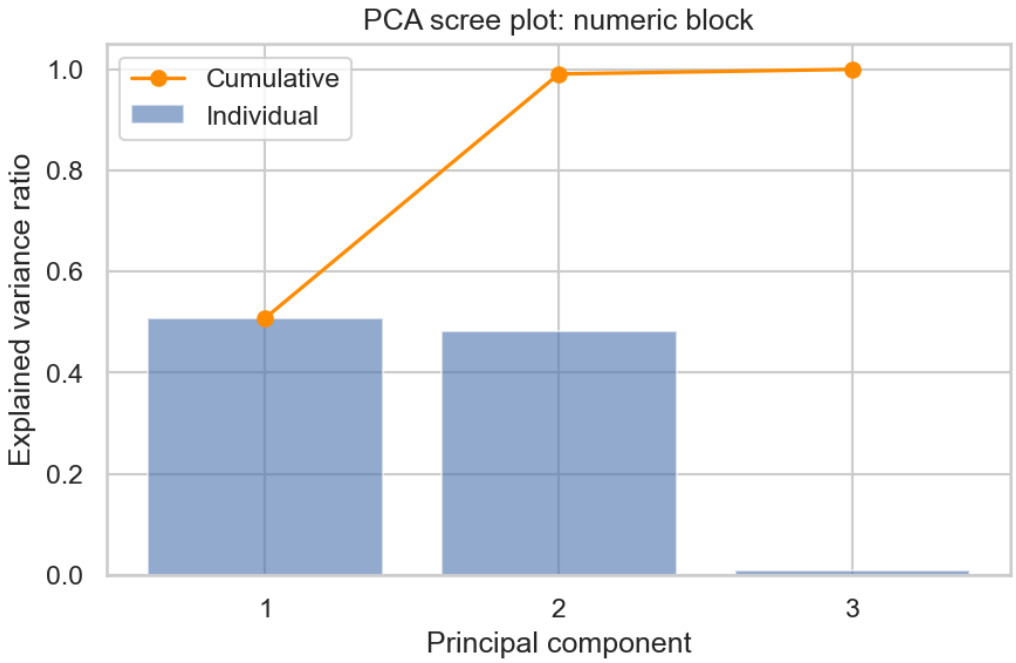


Figure 11. PCA scree plot — numeric block. All three components needed to explain ~100% of variance; no useful dimensionality reduction.

Bibliography

1. World Bank. *Remittance Prices Worldwide Database*. https://remittanceprices.worldbank.org/en/database
2. United Nations. *Sustainable Development Goal 10, Target 10.c*. https://sdgs.un.org/goals/goal10
3. Mei, Yajun. *ISyE 7406 — Data Mining & Statistical Learning*. Georgia Institute of Technology, Fall 2026.